

	A	B	C	D	E	F	G
1 Input data							
2 Initial inventory		5000					
3 Holding cost as % of prod cost		5%					
5 Month	1	2	3	4	5	6	
6 Production cost/unit	\$12.50	\$12.55	\$12.70	\$12.80	\$12.85	\$12.95	
7 Holding cost/unit	\$0.625	\$0.628	\$0.635	\$0.640	\$0.643	\$0.648	
8 Production plan (all quantities are in 100s of footballs)							
9 Month	1	2	3	4	5	6	
10 Units produced	5000	20000	30000	30000	25000	10000	
11	<=	<=	<=	<=	<=	<=	
12 Production capacity	30000	30000	30000	30000	30000	30000	
16 Demand	10000	15000	30000	35000	25000	10000	
18 Ending inventory (>=0)	0	5000	5000	0	0	0	
19	<=	<=	<=	<=	<=	<=	
20 Storage capacity	10000	10000	10000	10000	10000	10000	
22 Summary of costs (all costs are in hundreds of dollars)							
23 Month	1	2	3	4	5	6	Totals
24 Holding cost	0	3137.5	3175	0	0	0	6,312.50
25 Production cost	62500	251000	381000	384000	321250	129500	1,539,250.00
26							1,535,562.50

Input parameters

- n = number of decision periods
- p_t = unit production cost in period t
- h_t = unit holding cost in period t
- d_t = demand in period t
- κ = warehouse capacity
- u = production capacity
- l_0 = initial inventory

Decision variable

- x_t = production quantity in period t
- I_t = inventory at end of period t

Minimize

$$\sum_{t=1}^n (p_t x_t + h_t I_t)$$

Subject to the Constraints:

$$I_t = I_{t-1} + x_t - d_t \text{ for } t = 2, \dots, n$$

$$I_1 = I_0 + x_1 - d_1$$

$$0 \leq I_t \leq \kappa \text{ for } t = 1, 2, \dots, n$$

$$0 \leq x_t \leq u \text{ for } t = 1, 2, \dots, n$$

	A	B	C	D	E	F
1	x1	x2	x3	x4	x5	
2 fix cost	1500	1200	1600	1500	1600	
3 labor hour	2	1	6	4	8	
4 cloth use	3	2.5	4	4.5	5.5	
5 sell price	35	40	65	70	110	
6 cost price	20	10	25	30	35	
7 revenue	15	30	40	40	75	
9	x1	x2	x3	x4	x5	
10 produce?	0	1	0	0	1	
11 # produce	0	965.52	0	0	379.31	
12	<=	<=	<=	<=	<=	
13	0	1000000	0	0	1000000	

Decision variable

- Let $y_j = \begin{cases} 1 & \text{we produce type } j \\ 0 & \text{otherwise} \end{cases}$
- Let x_j = Number of types- j units to be produced; $j = 1, 2, \dots, 5$

Objective function

$$\text{Max } [(35 - 20)x_1 - 1500y_1] + \dots + [(110 - 35)x_5 - 1600y_5]$$

Constraints

$$2x_1 + \dots + 8x_5 \leq 4000$$

$$3x_1 + \dots + 5.5x_5 \leq 4500$$

$$x_j \geq 0; j = 1, 2, \dots, 5$$

$$y_j \in \{0, 1\}; j = 1, 2, \dots, 5$$

Set Objective: \$1\$7

To: ☒ Max ☐ Min ☐ Value Of:

By Changing Variable Cells:

\$B\$10:\$F\$11

Subject to the Constraints:

\$B\$10:\$F\$10 <= \$B\$13:\$F\$13

\$B\$11:\$F\$11 <= \$B\$13:\$F\$13

\$B\$11:\$F\$12 <= \$B\$13:\$F\$12

Problem statement

Revisit the multi-period production problem over an infinite planning horizon.

- Demand rate is constant at 20,000 units/month or $d = (12)(20000) = 240,000$ units/yr. $\rightarrow d = 240000$
- Fixed setup cost $K = 15000$ THB/batch $\rightarrow K = 15000$
- Unit holding cost 0.64 THB/unit/month or $h = (12)(0.64) = 7.68$ THB/unit/year. $\rightarrow h = 7.68$
- What batch size Q minimizes the average annual cost?

Model formulation

We formulate the classical economic order quantity (EOQ) model as a simple nonlinear optimization problem with a single decision variable.

Single item

demand rate, d (unit/year)	240000
Setup cost, K (THB/order)	15000
Unit holding cost, h (THB/unit/year)	7.68
Q	60000
Order frequency (order/year), d/Q	4
Avg inv. (unit), $Q/2$	30000
Avg setup cost (THB/year), Kd/Q	60000
Avg holding cost (THB/year), $hQ/2$	230400
Avg total cost (THB/year), $Kd/Q + hQ/2$	290400

Inputs

$a_{ij} = 1$ if candidate site j can cover demands at node i , and $a_{ij} = 0$ otherwise

f_j = fixed cost of locating a facility at candidate site j

Decision variable

$x_j \in \{0, 1\}$ if hub is located at j otherwise

Minimize

$$\sum_j f_j x_j$$

Subject to:

- (No. selected) $x_j + x_k + \dots + x_l = 1$ (Node A covered)
- (Node A covered) $x_j + x_k + \dots + x_l = 1$ (Node B covered)
- (Node B covered) $x_j + x_k + \dots + x_l = 1$ (Node C covered)
- (Node C covered) $x_j + x_k + \dots + x_l = 1$ (Node D covered)
- (Node D covered) $x_j + x_k + \dots + x_l = 1$ (Node E covered)
- (Node E covered) $x_j + x_k + \dots + x_l = 1$ (Node F covered)
- (Node F covered) $x_j + x_k + \dots + x_l = 1$ (Node G covered)
- (Node G covered) $x_j + x_k + \dots + x_l = 1$ (Node H covered)
- (Node H covered) $x_j + x_k + \dots + x_l = 1$ (Node I covered)
- (Node I covered) $x_j + x_k + \dots + x_l = 1$ (Node J covered)
- (Node J covered) $x_j + x_k + \dots + x_l = 1$ (Node K covered)
- (Node K covered) $x_j + x_k + \dots + x_l = 1$ (Node L covered)
- (Node L covered) $x_j + x_k + \dots + x_l = 1$ (Node M covered)
- (Node M covered) $x_j + x_k + \dots + x_l = 1$ (Node N covered)
- (Node N covered) $x_j + x_k + \dots + x_l = 1$ (Node O covered)
- (Node O covered) $x_j + x_k + \dots + x_l = 1$ (Node P covered)
- (Node P covered) $x_j + x_k + \dots + x_l = 1$ (Node Q covered)
- (Node Q covered) $x_j + x_k + \dots + x_l = 1$ (Node R covered)
- (Node R covered) $x_j + x_k + \dots + x_l = 1$ (Node S covered)
- (Node S covered) $x_j + x_k + \dots + x_l = 1$ (Node T covered)
- (Node T covered) $x_j + x_k + \dots + x_l = 1$ (Node U covered)
- (Node U covered) $x_j + x_k + \dots + x_l = 1$ (Node V covered)
- (Node V covered) $x_j + x_k + \dots + x_l = 1$ (Node W covered)
- (Node W covered) $x_j + x_k + \dots + x_l = 1$ (Node X covered)
- (Node X covered) $x_j + x_k + \dots + x_l = 1$ (Node Y covered)
- (Node Y covered) $x_j + x_k + \dots + x_l = 1$ (Node Z covered)

	A	B	C	D	E	F	G	H	I	J
1										
2	decis var	0	0	0	1	0	1	2		
3		1	1	0	1	0	0	1	>=	1
4		1	1	0	1	0	0	1	>=	1
5		0	0	1	0	1	1	1	>=	1
6		1	1	0	1	1	1	0	>=	1
7		0	0	1	1	1	1	0	>=	1
8		0	0	1	1	1	1	0	>=	1
9		0	0	1	0	0	1	1	>=	1
10		1	1	0	1	1	1	1	>=	1
11		0	0	1	1	1	1	0	>=	1
12		0	0	1	0	0	1	1	>=	1
13		1	1	1	1	1	1	1	>=	1
14		1	1	1	1	1	1	1	>=	1
15		1	1	1	1	1	1	1	>=	1
16		1	1	1	1	1	1	1	>=	1
17		1	1	1	1	1	1	1	>=	1
18		1	1	1	1	1	1	1	>=	1
19		1	1	1	1	1	1	1	>=	1
20		1	1	1	1	1	1	1	>=	1
21		1	1	1	1	1	1	1	>=	1
22		1	1	1	1	1	1	1	>=	1
23		1	1	1	1	1	1	1	>=	1
24		1	1	1	1	1	1	1	>=	1
25		1	1	1	1	1	1	1	>=	1
26		1	1	1	1	1	1	1	>=	1

Set Objective: \$H\$2

To: ☐ Max ☒ Min ☐ Value Of:

By Changing Variable Cells:

\$B\$2:\$G\$2

Subject to the Constraints:

\$B\$2:\$G\$2 <= \$H\$2:\$H\$2

\$H\$2:\$H\$2 <= \$I\$2:\$I\$2

\$I\$2:\$I\$2 <= \$J\$2:\$J\$2

\$J\$2:\$J\$2 <= \$K\$2:\$K\$2

\$K\$2:\$K\$2 <= \$L\$2:\$L\$2

\$L\$2:\$L\$2 <= \$M\$2:\$M\$2

\$M\$2:\$M\$2 <= \$N\$2:\$N\$2

\$N\$2:\$N\$2 <= \$O\$2:\$O\$2

\$O\$2:\$O\$2 <= \$P\$2:\$P\$2

\$P\$2:\$P\$2 <= \$Q\$2:\$Q\$2

\$Q\$2:\$Q\$2 <= \$R\$2:\$R\$2

\$R\$2:\$R\$2 <= \$S\$2:\$S\$2

\$S\$2:\$S\$2 <= \$T\$2:\$T\$2

\$T\$2:\$T\$2 <= \$U\$2:\$U\$2

\$U\$2:\$U\$2 <= \$V\$2:\$V\$2

\$V\$2:\$V\$2 <= \$W\$2:\$W\$2

\$W\$2:\$W\$2 <= \$X\$2:\$X\$2

\$X\$2:\$X\$2 <= \$Y\$2:\$Y\$2

\$Y\$2:\$Y\$2 <= \$Z\$2:\$Z\$2

\$Z\$2:\$Z\$2 <= \$AA\$2:\$AA\$2

\$AA\$2:\$AA\$2 <= \$AB\$2:\$AB\$2

\$AB\$2:\$AB\$2 <= \$AC\$2:\$AC\$2

\$AC\$2:\$AC\$2 <= \$AD\$2:\$AD\$2

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\$FS\$2:\$FS\$2 <= \$FT\$2:\$FT\$2

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\$FU\$2:\$FU\$2 <= \$FV\$2:\$FV\$2

\$FV\$2:\$FV\$2 <= \$FW\$2:\$FW\$2

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\$FX\$2:\$FX\$2 <= \$FY\$2:\$FY\$2

\$FY\$2:\$FY\$2 <= \$FZ\$2:\$FZ\$2

\$FZ\$2:\$FZ\$2 <= \$GA\$2:\$GA\$2

\$GA\$2:\$GA\$2 <= \$GB\$2:\$GB\$2

\$GB\$2:\$GB\$2 <= \$GC\$2:\$GC\$2

\$GC\$2:\$GC\$2 <= \$GD\$2:\$GD\$2

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\$GG\$2:\$GG\$2 <= \$GH\$2:\$GH\$2

\$GH\$2:\$GH\$2 <= \$GI\$2:\$GI\$2

\$GI\$2:\$GI\$2 <= \$GJ\$2:\$GJ\$2

\$GJ\$2:\$GJ\$2 <= \$GK\$2:\$GK\$2

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